

TITLE PAGE

Lost & Found Item Tracker

TraceBack — an AI-assisted platform that reconnects people with their belongings

Problem Statement ID	PCE SW PS 13 / SIH25	Theme	Smart Automation / Miscellaneous
Problem Statement Title	Lost & Found Item Tracker		
PS Category	Software	Team ID	to be filled
Team Name	to be filled	AISHE Code	to be filled

Fill the dashed fields with your team's details before submitting.

1 Friendly onboarding

Email + password registration with user / admin roles. Reporter identities are masked by default for privacy.

2 Dashboard — one-stop view

Stat cards (lost, found, active matches, returned), recent activity with live status badges, and top match previews as glowing confidence rings.

3 Report Lost / Found

A 3-step wizard (Details › Photo › Review) with a live preview card. Found reports add **private ownership questions** only the true owner can answer.

4 Smart matching engine

Compares every new report against all open opposite reports and produces a single, explainable confidence score with a per-factor breakdown.

5 Search & browse

Filter by category, location, date range and status; masked reporters keep listings privacy-safe.

6 Claim & verification

A step wizard: Submitted › Verification › Review › Handover › Returned. Answers are auto-scored; a human approves.

7 Secure messaging

Item-scoped chat so owner and finder coordinate a handover without sharing phone numbers.

8 Notifications

Match alerts, claim updates and messages in a live bell panel and a full page.

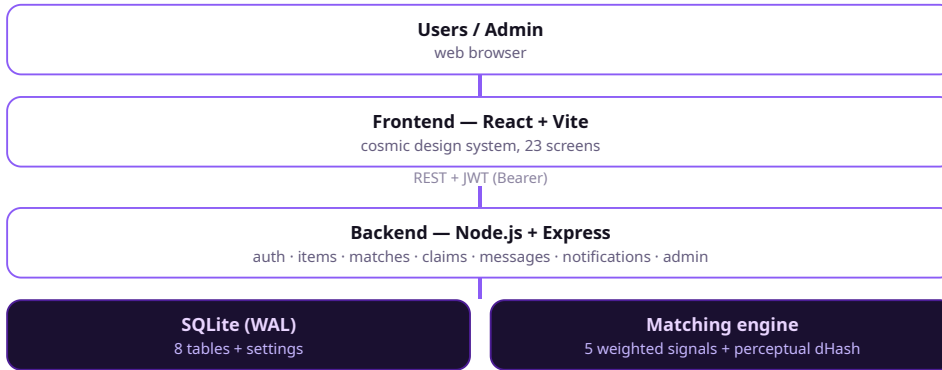
9 Admin console

Moderate users, items and claims; resolve disputes; view analytics; tune the matching weights live.

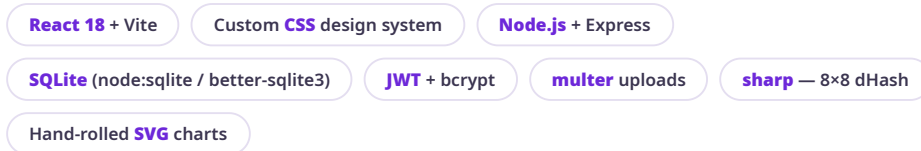
AI-driven decision support

- Weighted lost-vs-found match scoring across five signals.
- Image-based similarity from farmer/finder-uploaded photos (perceptual hash).
- Advisory ownership-verification scoring from private answers.
- Analytics: match success rate, resolution time, category & location hotspots.

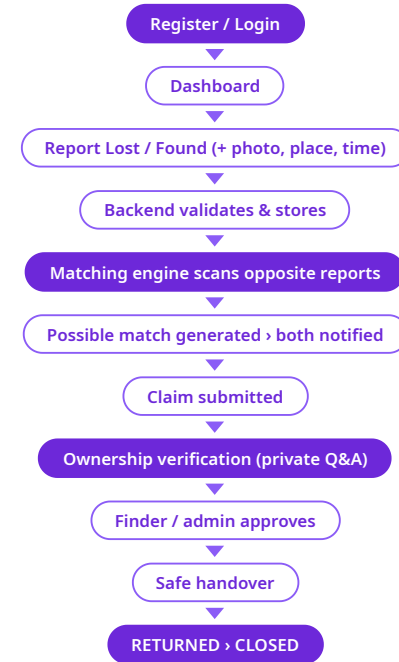
• Architecture



• Technical stack



• Workflow



● Weighted signals (never relies on an exact name)

Item / category similarity		25%
Description similarity		25%
Location similarity		20%
Date / time proximity		15%
Image similarity (dHash)		15%

● Techniques

- **Synonym lexicon** — phone>mobile>cell, wallet>purse, canteen>cafeteria>mess, colour & place families.
- **Text similarity** — token overlap + Jaccard + character-bigram cosine (tolerates typos & different wording).
- **Calibration curve** — same-object descriptions share only 30–45% of words; mapped to real confidence.
- **Date proximity** — exponential decay across a configurable window.
- **Image similarity** — 64-bit perceptual difference hash, Hamming distance.
- **Weight re-normalisation** — a missing factor (no photo) is dropped and the rest rescale.

Worked example — from the live demo

Black wallet, same canteen, same afternoon, similar description:

Category & item name	81%
Description	75%
Location	100%
Date / time	100%
Image	94%
Combined score	88% — Possible Match

Ownership verification

- Finder stores private questions; the API returns only the questions, never the answers.
- Claimant answers are auto-scored by the same engine — advisory only.
- Genuine owner ~**89%** vs impostor ~**2%** in testing; a human always decides.
- Rejections can be escalated as a dispute to an admin.

● Feasibility

TraceBack is highly feasible — it is already a complete, working full-stack build, verified end to end (44 automated checks over the whole lost>found>match>return journey).

Technical

Standard, well-supported stack (Node, React, SQLite). Runs on any laptop with zero database setup and no C++ toolchain — the DB auto-uses Node's built-in SQLite.

Operational

Simple web UI any student can use; the matching, verification and status tracking are automatic. A single command boots the whole app.

Financial

Near-zero running cost: SQLite + local image hashing need no paid APIs. A low-cost MVP is already done.

Deployable now

Single Node process serves the built client; hostable on any free/low-cost PaaS or a campus server.

● Viability

Real, recurring need

Every campus, office and transit hub loses items daily; the current process is scattered notice boards and chat groups.

Privacy-first trust model

Masked identities + private ownership questions make institutions comfortable adopting it officially.

Modular & scalable design

Plain-SQL data layer swaps to PostgreSQL for multi-campus; the matching engine is an isolated module ready for heavier ML.

Adoption paths

Deploy per college / company; integrate with existing ID systems; extend to city-level public lost-and-found.

Sustainability

Institutional licensing, optional premium notifications, and analytics for facilities teams — with no dependency on paid third-party services.

Faster

REUNIONS VS NOTICE BOARDS

Higher

RETURN RATE VIA MATCHING

Safer

VERIFIED OWNERSHIP

Zero

PAID-API DEPENDENCY

● Impact**1****Reconnects people with belongings**

Turns a scattered, manual process into an automated, trackable one — items get back to owners that would otherwise be lost for good.

2**Digital transformation of lost-and-found**

Replaces WhatsApp groups and paper registers with a searchable, auditable system for the whole institution.

3**Builds trust & safety**

Verification and masked identities prevent false claims and protect users' contact details.

4**Reduces waste**

Returned items mean fewer replacements bought — a small but real sustainability win.

● Key benefits

- **Smart matching** — AI-assisted scoring surfaces likely pairs instantly, no manual searching.
- **Image similarity** — photos are compared perceptually, catching matches text alone would miss.
- **Ownership verification** — private questions ensure the right person gets the item.
- **Live status tracking** — everyone sees exactly where a case stands, end to end.
- **Secure in-app chat** — coordinate handovers without exposing phone numbers.
- **Admin analytics** — success rate, resolution time and hotspots for facilities teams.
- **Runs anywhere** — offline-capable dev setup, no paid services, one-command boot.

• Comparison with existing approaches

FEATURE	TRACEBACK	NOTICE BOARD / WHATSAPP GROUP	GENERIC CLASSIFIEDS	BASIC COLLEGE PORTAL
Automatic lost-found matching	Yes — 5-signal engine	No	No	No
Image-based similarity	Yes (perceptual hash)	No	No	No
Ownership verification	Yes (private Q&A + auto-score)	No	No	Rare
Masked identity / privacy	Yes, until approval	No (numbers public)	Partial	Partial
Status tracking & lifecycle	Yes (6 stages)	No	No	Basic
Secure in-app chat	Yes (item-scoped)	Group only	Varies	No
Search & filter	Yes (category/place/date/status)	No	Yes	Basic
Admin moderation & analytics	Yes	No	Limited	Limited
Cost to run	Near-zero, no paid APIs	Free but manual	Ads/fees	IT overhead

What makes TraceBack different

- It **actively matches** instead of just listing — the core gap in every existing option.
- It **verifies ownership** before handover, which listings and chats cannot.
- It is **privacy-first and auditable** — suitable for official institutional adoption.

References & resources

- Live source code & docs: GitHub repository (this project).
- UI walkthrough: [docs/SCREENSHOTS.md](#) (29 screenshots).
- Technical guide: [docs/TraceBack-SIH-Guide.pdf](#) .
- End-to-end test: [server/e2e-demo.mjs](#) (44 assertions).
- Add your deployed demo URL and demo video link here before submitting.